Reactive Following with Five Robots: A Five-Rule Controller in Webots and ROS 2

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***Abstract*—We present a purely reactive controller that makes five differential-drive robots form a following chain in Webots, driven by five ROS 2 nodes. Each robot acts on a single current snapshot from a frontal camera and LiDAR (100°, 4 m); there is no target memory, no global pose and no inter-robot communication. Robot orientation is read directly from colour: the front half of each chassis is red and the rear half green, so "facing me" and "facing away" reduce to which colour is nearer. The controller is five prioritised rules. In the reported run the five robots formed a stable train — one leader on its base curve and four followers — after 186 s, and held it for the remaining 105 s with no avoidance event of any kind. We report following duration, angular error and target losses, and devote the larger part of this paper to the quirks and failure modes that remain, several of which are consequences of the sensing envelope rather than of the control rules.**

I. Introduction

The task is reactive following: five robots, each deciding only from its own current sensor readings, should aggregate into one or more following chains. The constraints are those of classical reactive control in the sense of Brooks' subsumption architecture [1]: behaviour is a priority ordering over situation–action rules, with no world model and no planning. The assignment additionally forbids GPS, global localisation, target memory and privileged state, while permitting internal controller state of the PI/PID kind.

Our contribution is deliberately small. The interesting content of this report is not the controller, which is roughly 120 lines, but the catalogue of ways in which a correct-looking reactive rule set misbehaves once five copies of it interact in a shared environment. Section VI is therefore the longest section, by intent.

II. Simulation Model and Sensing

A. Robot

Each robot is a differential-drive cylinder of radius 0.115 m with two powered wheels and an invisible contact bumper of radius 0.146 m. Two robots are in contact when their centres are 0.292 m apart; this number is the reference against which all proximity results below are read.

B. Colour-coded orientation

An earlier iteration used two small recognition plaques, one red at the front and one green at the rear. These were replaced by two coloured half-shells covering the front and rear halves of the chassis. The reason is angular coverage: a flat plaque shrinks to nothing in projection as a robot turns away, and we measured that the leader became unclassifiable once the angle between its heading and the observer's line of sight exceeded about 74° — precisely when a leader was turning, which is when a follower most needs to keep it.

Each half is a separate Webots `Solid` with its own `recognitionColors`. One detail is load-bearing: Webots reports the *origin* of a recognised Solid, so the two halves are offset to $x = \pm 0.060\,\mathrm{m}$. Sharing the chassis centre would make both halves report the same position and front/back would be indistinguishable.

C. Sensors

Sensing is frontal only, as required: a recognition camera and a LiDAR, both 100° and 4 m. The LiDAR mounting deserves comment because it caused most of our early failures and its fix is non-obvious.

Mounted centrally on the roof at $z = 0.155\,\mathrm{m}$ — the natural place, and necessary to clear its own chassis — the LiDAR sits *above* the 0.125 m body top of every other robot. With a narrow vertical field of view, the beam only descends to body height at

$$d \;=\; \frac{0.155 - 0.125}{\tan(\theta_v/2)} \;\approx\; 0.75\,\mathrm{m}, \tag{1}$$

so a peer closer than 0.75 m is invisible. Every collision threshold we might wish to set lies inside that blind radius. Worse, the blindness is unavoidable for a *central* sensor: to avoid seeing its own rim at radius r_b the beam drop at r_b must stay under the clearance c, while seeing a peer at distance d requires the drop at $d - r_b$ to reach c. Both θ_v and c cancel, leaving $d > 2r_b$ — twice the body radius, which is exactly the distance at which two robots already touch.

The resolution exploits the fact that the LiDAR is now frontal. Mounted at the *front edge* ($x = 0.130\,\mathrm{m}$, outside the body radius), every forward ray's closest approach to the chassis lies behind the sensor, so it can be lowered to $z = 0.115\,\mathrm{m}$: inside the 0.045 m to 0.125 m band of every peer. Robots are then visible at all ranges including contact. A 360° LiDAR cannot do this, because its rearward rays would face its own chassis.

TABLE I
RULES, IN EVALUATION ORDER. FIRST MATCH WINS.

#	State	Condition	Action
1	`ESCAPE_TURN`	red nearer, $d_r < 0.45$	$v = 0$, turn right
2	`AVOID_RED`	red nearer, $d_r < 0.90$	turn right
3	`FOLLOW`	green visible	steer to it, speed ramp
4	`APPROACH`	red only, $d_r \geq 0.90$	steer to it
5	`AVOID_WALL`	no robot, LiDAR < 2.0	turn to open side
–	`BASE_CURVE`	otherwise	v_0, ω_{base}

TABLE II
PER-ROBOT RESULTS, 291.2 S RUN.

	Following	%	Losses	Mean $\lvert\beta\rvert$ (rad)
R1	290.9 s	99.9	2	0.629
R2	291.1 s	99.9	1	0.702
R3	44.7 s	15.3	2	0.420
R4	282.8 s	97.1	3	0.729
R5	137.2 s	47.1	6	0.451

D. Software architecture

Webots runs one controller per robot, which packs a camera/LiDAR snapshot into a UDP datagram. A ROS 2 bridge republishes each as `/Ri/local_sensors`; five independent `ReactiveFollower` nodes subscribe and publish `/Ri/cmd_vel`. No node subscribes to another robot's topics, which is what makes the no-communication constraint auditable rather than merely asserted.

III. REACTIVE CONTROLLER

Perception reduces each frame to the nearest red and the nearest green, as (distance, bearing) pairs. The controller is five rules in priority order (Table I).

Two design points proved necessary rather than cosmetic.

a) Red is a comparison, not a detection: Rules 1–2 require red to be *nearer than green*, not merely visible. Every robot being followed presents a red half somewhere in view, so keying on visibility made followers abandon correct leaders.

b) Turn rate is capped by radius, not by a fixed rate: Steering straight at a leader is pure pursuit, which cuts inside the leader's path. In a five-robot ring the leader sits about 0.63 rad off-axis, giving $\omega = 1.5 \times 0.63 = 0.94\,\text{rad}\,\text{s}^{-1}$ and an arc radius of $v/\omega \approx 0.15\,\text{m}$ against a 0.95 m ring: the formation spirals inward until it piles up. We cap by $\omega_{\max} = v_0/R_{\min}$ rather than by a shared ω, because the robots run at different speeds ($0.12\,\text{m}\,\text{s}^{-1}$ to $0.16\,\text{m}\,\text{s}^{-1}$) and a shared cap would give each a different radius and distort the ring.

IV. EXPERIMENTAL METHOD

The five robots start evenly spaced on a 0.95 m circle, each heading along the tangent. We record per robot: following duration, mean absolute bearing error while following, target losses (transitions out of a following state) and bumper contacts. A seeded script (`tools/randomize_starts.py`) regenerates scattered starting poses so that runs can be repeated with different positions and headings while remaining reproducible.

V. RESULTS

Table II reports a 291 s run. The controller issued only 48 mode changes across all five robots, i.e. $9.9\,\text{min}^{-1}$.

Time-weighted occupancy was `FOLLOW` 71.9%, `BASE_CURVE` 27.5%, and all avoidance states together below 1% (`AVOID_RED` 0.3%, `ESCAPE_TURN` 0.2%, `AVOID_WALL` 0.1%).

The run passed through five phases: 82 s of scattered search; 34 s of first acquisitions; an 11 s congestion incident; 59 s of re-forming; and then 105 s of stable train with no avoidance event at all. The end state is the intended one — R3 leading on its base curve, R1, R2, R4 and R5 following — which is why R3's low following percentage is a success rather than a failure: the leader of a train is, by definition, not following anything.

Mean absolute bearing error while following was 0.549 rad ($31.5°$). This should not be read as tracking error. In a five-robot ring the leader is geometrically about $36°$ off-axis, so the measured value is close to the best achievable for this formation.

VI. QUIRKS, SHORTCOMINGS AND LIMITATIONS

A. Edge-of-cone bistability

The one congestion incident is instructive. R3 and R5 each detected an oncoming robot, turned right and cleared in about 1.4 s: the rule working as designed. R4 instead oscillated between `AVOID_WALL` and `ESCAPE_TURN` three times in 4 s. The cause is visible in the bearing at resolution, $+49.2°$: its target sat on the $\pm 50°$ cone edge and flickered in and out of detection. The two states require opposite readings — "red very close" and "no robot at all" — so the controller alternated between them. This is a sensor-envelope artefact, not a logic error; no rule is wrong. A hysteresis band would suppress it at the cost of a small amount of remembered state.

B. The emergency response is rate-limited

An acceleration limiter smooths all commands. It also smooths the emergency: `ESCAPE_TURN` commands $v = 0$, $\omega = -0.80$, but the log shows $v = 0.12$, $\omega = +0.37$ at onset, i.e. the robot is still rolling forward and still turning *left* for roughly 0.2 s while the limiter unwinds the previous command. The highest-priority rule is therefore not actually the fastest-acting one.

C. Collisions were not captured

The metrics summary prints only on a clean shutdown. The analysed run was killed rather than interrupted, so bumper counts were lost. From mode thresholds we can infer that `ESCAPE_TURN` triggers at 0.64 m centre-to-centre against a 0.292 m contact distance, so the incident was clear at the trigger — but the logs of that run cannot say how much closer it came. Distance logging and a closest-approach metric were

added afterwards and are not reflected in the numbers above. We report this rather than quietly omitting the metric.

D. No rear or lateral sensing

A robot alongside or behind is invisible: the camera and LiDAR both span only 100° forward. Sideways collisions are therefore undetectable in principle, not merely unhandled. Our mitigations are preventive — lane geometry and turn-rate caps that stop robots ending up abreast — and nothing reacts once they are.

E. Open-loop geometric coupling

Three parameters are not independent. For five robots on a ring, spacing and radius satisfy $s = 2R\sin(\pi/5) = 1.176R$, and the camera reads 0.19 m less than centre spacing. An earlier standoff of 0.45 m therefore demanded a 0.54 m ring while the turn cap refused anything tighter than 0.90 m; the robots bunched and crawled at the speed floor. The present values agree, but they were made to agree by hand and nothing in the code enforces the relationship.

F. Mode changes are the only log events

Modes are logged on change, so a quantity that drifts *within* a mode is invisible. We initially misdiagnosed a collision because `APPROACH` was recorded at $\omega = 0.27$ while its true peak, reached between log lines, was at least 0.73. Any conclusion drawn from this log format understates within-state extremes.

G. Measurement artefacts

One earlier analysis attributed 70% of robot-time to sensor dropouts and concluded the simulation was running below real time. The log had in fact been recorded while stepping Webots tick by tick, so the simulation was halted between clicks. When run freely, `SENSOR_FAILSAFE` occupies 0.0% of the run. We record the mistake because the artefact is easy to reproduce and the wrong conclusion was superficially well-supported.

H. Repository state

The test suite does not match the delivered controller. All ten test modules exercise the previous `safe_logic` cascade and the marker-based perception module, neither of which is wired into the current system; they would not run against it. The old controller is retained in the repository for comparison but is not executed.

I. Single-run evidence

The results are one run. The seeded randomiser exists so that repeats with different initial conditions can be performed, and the comparison across turning strengths that the brief suggests (no following / weak / strong) is supported by a command-line flag, but neither has been run at the time of writing. No statistical claim is made.

VII. Conclusion

A five-rule reactive controller, acting on one frontal snapshot at a time, assembled five robots into a stable leader-plus-four-followers train in 186 s and held it without further intervention. The rules were cheap; the difficulty was almost entirely in the interaction between the sensing envelope and the geometry of a multi-robot formation. The dominant failure mode we did not eliminate — bistable detection at the edge of the field of view — is a property of where the sensor stops, not of what the controller decides.

References

[1] R. A. Brooks, "A robust layered control system for a mobile robot," *IEEE J. Robot. Autom.*, vol. 2, no. 1, pp. 14–23, 1986.

[2] R. C. Arkin, *Behavior-Based Robotics*. Cambridge, MA, USA: MIT Press, 1998.

[3] O. Michel, "Webots: Professional mobile robot simulation," *Int. J. Adv. Robot. Syst.*, vol. 1, no. 1, pp. 39–42, 2004.

[4] S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, "Robot Operating System 2: Design, architecture, and uses in the wild," *Science Robotics*, vol. 7, no. 66, 2022.

[5] R. C. Coulter, "Implementation of the pure pursuit path tracking algorithm," Robotics Institute, Carnegie Mellon Univ., Tech. Rep. CMU-RI-TR-92-01, 1992.